

Solutions to Quantum Mechanics: Concepts and
Applications
by Nouredine Zettili

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*To Dr. Miller Mendoza Jiménez from ETH Zürich
Thank you for inspiring me to do theoretical physics.*

Preface

This is an ongoing project of mine to solve a significant percentage of all the problems in the third edition of Zettili's *Quantum Mechanics* [Zet22]. I frequently refer to [Axl24] as linear algebra is embedded in quantum theory. As always, should there be any errors, please email me:

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The only way to understand physics is by *doing* physics. However, working your way through problems is useless without the ability to verify your solutions. There is good reason why books on the subject (usually) do not make the answers public, for immediately looking at the answers the moment you get stuck on a problem is extremely detrimental to one's comprehension of a topic. That said, it is also detrimental to an individual's understanding if they cannot verify their workings. Therefore, it is imperative that you, the reader, use this manual *wisely* and do not give up on a difficult problem. Of course, if you really can't solve something, do consult the solutions, but always remember that the lesson of an exercise is not the exercise itself, but rather the way it forces you to organise your thoughts before solving it.

I hope this manuscript becomes useful to the theoretical physics community (undergraduate students) as I continue to update it with more and more solutions.

Kieron Eisma

ACKNOWLEDGMENTS AND AI DISCLOSURE

ChatGPT 5.5 has been used to partially typeset this document (converting my working into TeX code), and to generate images to aid readers' visualisation. All solutions, however, have been completely fabricated by myself.

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Chapter 1

Origins of Quantum Physics

Exercise 1.37.

Consider the function

$$\psi(x) = \begin{cases} 1 - |x|, & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

To find the Fourier Transform $\phi(p)$ of this function, consult Exercise 2 on page 161 of [SS03].

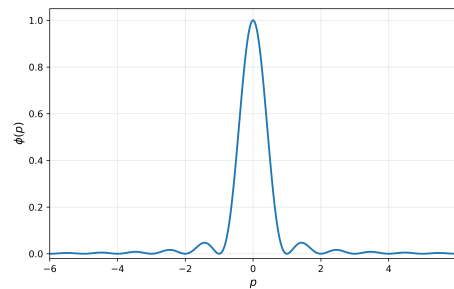


Figure 1.1: *Plot of the Fourier transform $\phi(p)$ of ψ . Plot generated by ChatGPT 5.5*

Chapter 2

Mathematical Tools of Quantum Mechanics

Exercise 2.1(a)

For the first two norms, we have:

$$\begin{aligned}\langle\psi|\psi\rangle &= (-i\langle\phi_1| - 3i\langle\phi_2| - \langle\phi_3|)(i|\phi_1\rangle + 3i|\phi_2\rangle - |\phi_3\rangle) \\ &= (-i)(i)\langle\phi_1|\phi_1\rangle + (-i)(3i)\langle\phi_1|\phi_2\rangle + (-i)(-1)\langle\phi_1|\phi_3\rangle \\ &\quad + (-3i)(i)\langle\phi_2|\phi_1\rangle + (-3i)(3i)\langle\phi_2|\phi_2\rangle + (-3i)(-1)\langle\phi_2|\phi_3\rangle \\ &\quad + (-1)(i)\langle\phi_3|\phi_1\rangle + (-1)(3i)\langle\phi_3|\phi_2\rangle + (-1)(-1)\langle\phi_3|\phi_3\rangle \\ &= 1 + 9 + 1 \\ &= 11,\end{aligned}$$

and

$$\begin{aligned}\langle\chi|\chi\rangle &= (\langle\phi_1| + i\langle\phi_2| - 5i\langle\phi_3|)(|\phi_1\rangle - i|\phi_2\rangle + 5i|\phi_3\rangle) \\ &= \langle\phi_1|\phi_1\rangle + \langle\phi_2|\phi_2\rangle + 25\langle\phi_3|\phi_3\rangle \\ &= 1 + 1 + 25 \\ &= 27.\end{aligned}$$

For a much faster method to calculate these norms, see 6.30 (b) in [Axl24]

Now, for $\langle\chi|\psi\rangle$ and $\langle\psi|\chi\rangle$:

$$\begin{aligned}
\langle \chi | \psi \rangle &= (\langle \phi_1 | + i\langle \phi_2 | - 5i\langle \phi_3 |)(i|\phi_1\rangle + 3i|\phi_2\rangle - |\phi_3\rangle) \\
&= i\langle \phi_1 | \phi_1 \rangle - 3\langle \phi_2 | \phi_2 \rangle + 5i\langle \phi_3 | \phi_3 \rangle \\
&= i - 3 + 5i \\
&= -3 + 6i,
\end{aligned}$$

and

$$\begin{aligned}
\langle \psi | \chi \rangle &= (-i\langle \phi_1 | - 3i\langle \phi_2 | - \langle \phi_3 |)(|\phi_1\rangle - i|\phi_2\rangle + 5i|\phi_3\rangle) \\
&= -i\langle \phi_1 | \phi_1 \rangle - 3\langle \phi_2 | \phi_2 \rangle - 5i\langle \phi_3 | \phi_3 \rangle \\
&= -i - 3 - 5i \\
&= -3 - 6i.
\end{aligned}$$

Note that the line of working for $\langle \psi | \chi \rangle$ is redundant because we could have taken the complex conjugate of $\langle \chi | \psi \rangle$.

As expected, we have that $\langle \chi | \psi \rangle = \langle \psi | \chi \rangle^*$.

For $\langle \psi + \chi | \psi + \chi \rangle$:

$$\begin{aligned}
\langle \psi + \chi | \psi + \chi \rangle &= \langle \psi | \psi \rangle + \langle \psi | \chi \rangle + \langle \chi | \psi \rangle + \langle \chi | \chi \rangle \\
&= 11 + (-3 - 6i) + (-3 + 6i) + 27 \\
&= 32.
\end{aligned}$$

Exercise 2.1(b)

To compute the operators $|\psi\rangle\langle\chi|$ and $|\chi\rangle\langle\psi|$:

$$\begin{aligned}
|\psi\rangle\langle\chi| &= (i|\phi_1\rangle + 3i|\phi_2\rangle - |\phi_3\rangle)(\langle\phi_1| + i\langle\phi_2| - 5i\langle\phi_3|) \\
&= i|\phi_1\rangle\langle\phi_1| + i^2|\phi_1\rangle\langle\phi_2| - 5i^2|\phi_1\rangle\langle\phi_3| \\
&\quad + 3i|\phi_2\rangle\langle\phi_1| + 3i^2|\phi_2\rangle\langle\phi_2| - 15i^2|\phi_2\rangle\langle\phi_3| \\
&\quad - |\phi_3\rangle\langle\phi_1| - i|\phi_3\rangle\langle\phi_2| + 5i|\phi_3\rangle\langle\phi_3| \\
&= i|\phi_1\rangle\langle\phi_1| - |\phi_1\rangle\langle\phi_2| + 5|\phi_1\rangle\langle\phi_3| \\
&\quad + 3i|\phi_2\rangle\langle\phi_1| - 3|\phi_2\rangle\langle\phi_2| + 15|\phi_2\rangle\langle\phi_3| \\
&\quad - |\phi_3\rangle\langle\phi_1| - i|\phi_3\rangle\langle\phi_2| + 5i|\phi_3\rangle\langle\phi_3|,
\end{aligned}$$

and

$$\begin{aligned}
|\chi\rangle\langle\psi| &= (|\phi_1\rangle - i|\phi_2\rangle + 5i|\phi_3\rangle)(-i\langle\phi_1| - 3i\langle\phi_2| - \langle\phi_3|) \\
&= -i|\phi_1\rangle\langle\phi_1| - 3i|\phi_1\rangle\langle\phi_2| - |\phi_1\rangle\langle\phi_3| \\
&\quad - i(-i)|\phi_2\rangle\langle\phi_1| - 3i(-i)|\phi_2\rangle\langle\phi_2| + i|\phi_2\rangle\langle\phi_3| \\
&\quad - 5i^2|\phi_3\rangle\langle\phi_1| - 15i^2|\phi_3\rangle\langle\phi_2| - 5i|\phi_3\rangle\langle\phi_3| \\
&= -i|\phi_1\rangle\langle\phi_1| - 3i|\phi_1\rangle\langle\phi_2| - |\phi_1\rangle\langle\phi_3| \\
&\quad - |\phi_2\rangle\langle\phi_1| - 3|\phi_2\rangle\langle\phi_2| + i|\phi_2\rangle\langle\phi_3| \\
&\quad + 5|\phi_3\rangle\langle\phi_1| + 15|\phi_3\rangle\langle\phi_2| - 5i|\phi_3\rangle\langle\phi_3|.
\end{aligned}$$

The traces are easy to compute, because $\text{Tr}(\hat{A}\hat{B}) = \text{Tr}(\hat{B}\hat{A})$ (for a proof, see 8.56 in [Axl24]):

$$\text{Tr}(|\psi\rangle\langle\chi|) = \text{Tr}(\langle\chi|\psi\rangle) = -3 + 6i$$

$$\text{Tr}(|\chi\rangle\langle\psi|) = \text{Tr}(\langle\psi|\chi\rangle) = -3 - 6i$$

Notice how the traces are complex conjugates of one another.

Exercise 2.1 (c)

The computation of the requested Hermitian conjugates of the quantum states are trivial.

Exercise 2.2

Consider the state vector $|\psi_1\rangle = |\phi_1\rangle + 4i|\phi_2\rangle + 5|\phi_3\rangle$. Taking complex conjugates and interchanging kets with bras, we obtain

$$\langle\psi_1| = \langle\phi_1| - 4i\langle\phi_2| + 5\langle\phi_3|,$$

Since $|\psi_2\rangle = b|\phi_1\rangle + 4|\phi_2\rangle - 3i|\phi_3\rangle$, we compute the inner product

$$\begin{aligned}
\langle\psi_1|\psi_2\rangle &= (\langle\phi_1| - 4i\langle\phi_2| + 5\langle\phi_3|)(b|\phi_1\rangle + 4|\phi_2\rangle - 3i|\phi_3\rangle) \\
&= b\langle\phi_1|\phi_1\rangle + 4\langle\phi_1|\phi_2\rangle - 3i\langle\phi_1|\phi_3\rangle \\
&\quad - 4ib\langle\phi_2|\phi_1\rangle - 16i\langle\phi_2|\phi_2\rangle + 12\langle\phi_2|\phi_3\rangle \\
&\quad + 5b\langle\phi_3|\phi_1\rangle + 20\langle\phi_3|\phi_2\rangle - 15i\langle\phi_3|\phi_3\rangle \\
&= b - 16i - 15i \\
&= b - 31i.
\end{aligned}$$

For the quantum states to be orthogonal, require that $\langle \psi_1 | \psi_2 \rangle := 0$, which forces $b = 31i$ by the equation above.

Again, for a much quicker calculation, see 6.30(c) in [Axl24].

Exercise 2.3

All the necessary products for you to verify (a) the Cauchy-Schwarz Inequality and (b) the Triangle Inequality are contained in Exercise 2.1(a).

These inequalities must always be true, by 6.14 and 6.17 in [Axl24], so the result is no surprise.

Exercise 2.4

Consider the quantum states

$$\begin{aligned} |\psi\rangle &= \alpha|\phi_1\rangle + 5|\phi_2\rangle \\ |\chi\rangle &= 3\alpha|\phi_1\rangle - 4|\phi_2\rangle. \end{aligned}$$

Taking the Hermitian conjugate of the state $|\psi\rangle$ yields the bra

$$\langle\psi| = \bar{\alpha}\langle\phi_1| + 5\langle\phi_2|,$$

so that

$$\begin{aligned} \langle\psi|\chi\rangle &= (\bar{\alpha}\langle\phi_1| + 5\langle\phi_2|)(3\alpha|\phi_1\rangle - 4|\phi_2\rangle) \\ &= 3\alpha\bar{\alpha}\langle\phi_1|\phi_1\rangle - 4\bar{\alpha}\langle\phi_1|\phi_2\rangle \\ &\quad + 15\alpha\langle\phi_2|\phi_1\rangle - 20\langle\phi_2|\phi_2\rangle \\ &= 3|\alpha|^2 - 20. \end{aligned}$$

In order for the states $|\psi\rangle$ and $|\chi\rangle$ to be orthogonal, mandate

$$\langle\psi|\chi\rangle = 0,$$

which forces

$$|\alpha|^2 = \frac{20}{3}.$$

Now for $\alpha \in \mathbb{R}$, we can either have $\alpha = \sqrt{\frac{20}{3}}$ or $\alpha = -\sqrt{\frac{20}{3}}$.

More generally, observe that, if $B(x, r)$ denotes the open ball of radius r around a point x in a metric space (X, d) , we see that for all

$$\alpha \in \partial B\left(\mathbf{0}_{\mathbb{C}}, \sqrt{20/3}\right),$$

the states $|\psi\rangle$ and $|\chi\rangle$ will be orthonormal.

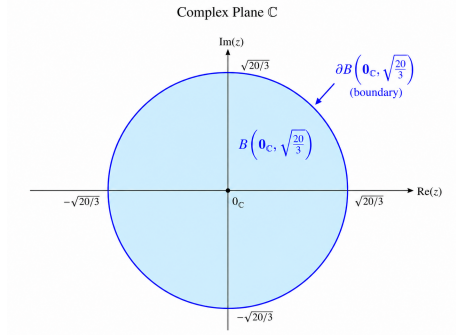


Figure 2.1: The complex ball centered about the origin (image generated by ChatGPT)

Exercise 2.5

Consider the quantum states $|\psi\rangle = |\phi_1\rangle + |\phi_2\rangle$ and $|\chi\rangle = |\phi_1\rangle - |\phi_2\rangle$, where $|\phi_1\rangle$ and $|\phi_2\rangle$ are *not* orthonormal. We claim that we have the following identities:

$$\begin{aligned}\langle\psi|\psi\rangle + \langle\chi|\chi\rangle &= 2\langle\phi_1|\phi_1\rangle + 2\langle\phi_2|\phi_2\rangle, \\ \langle\psi|\psi\rangle - \langle\chi|\chi\rangle &= 2\langle\phi_1|\phi_2\rangle + 2\langle\phi_2|\phi_1\rangle.\end{aligned}$$

Proof:

Taking the Hermitian conjugate of the state vector $|\psi\rangle = |\phi_1\rangle + |\phi_2\rangle$ yields the bra $\langle\psi| = \langle\phi_1| + \langle\phi_2|$. Consequently, the norm of $|\psi\rangle$ is given by

$$\begin{aligned}\langle\psi|\psi\rangle &= (\langle\phi_1| + \langle\phi_2|)(|\phi_1\rangle + |\phi_2\rangle) \\ &= \langle\phi_1|\phi_1\rangle + \langle\phi_1|\phi_2\rangle + \langle\phi_2|\phi_1\rangle + \langle\phi_2|\phi_2\rangle.\end{aligned}$$

Similarly, we have $\langle\chi| = \langle\phi_1| - \langle\phi_2|$,

which yields

$$\begin{aligned}\langle\chi|\chi\rangle &= (\langle\phi_1| - \langle\phi_2|)(|\phi_1\rangle - |\phi_2\rangle) \\ &= \langle\phi_1|\phi_1\rangle - \langle\phi_1|\phi_2\rangle - \langle\phi_2|\phi_1\rangle + \langle\phi_2|\phi_2\rangle.\end{aligned}$$

It immediately follows that

$$\begin{aligned}
\langle \psi | \psi \rangle + \langle \chi | \chi \rangle &= (\langle \phi_1 | \phi_1 \rangle + \langle \phi_1 | \phi_2 \rangle + \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&\quad + (\langle \phi_1 | \phi_1 \rangle - \langle \phi_1 | \phi_2 \rangle - \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&= 2\langle \phi_1 | \phi_1 \rangle + 2\langle \phi_2 | \phi_2 \rangle,
\end{aligned}$$

and

$$\begin{aligned}
\langle \psi | \psi \rangle - \langle \chi | \chi \rangle &= (\langle \phi_1 | \phi_1 \rangle + \langle \phi_1 | \phi_2 \rangle + \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&\quad - (\langle \phi_1 | \phi_1 \rangle - \langle \phi_1 | \phi_2 \rangle - \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&= 2\langle \phi_1 | \phi_2 \rangle + 2\langle \phi_2 | \phi_1 \rangle,
\end{aligned}$$

which is the desired result. ■

Exercise 2.6(a)

Consider the quantum state

$$|\psi\rangle = \frac{1}{\sqrt{15}}|\phi_1\rangle + \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle$$

where the kets $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$ are orthonormal.

By orthonormality, we have that

$$\begin{aligned}
\langle \psi | \psi \rangle &= \left(\frac{1}{\sqrt{15}} \right)^2 + \left(\frac{1}{\sqrt{3}} \right)^2 + \left(\frac{1}{\sqrt{5}} \right)^2 \\
&= \frac{1}{15} + \frac{1}{3} + \frac{1}{5} \\
&= \frac{9}{15} \\
&= \frac{3}{5}.
\end{aligned}$$

It follows the norm of $|\psi\rangle$ is given by

$$\|\psi\| = \sqrt{\frac{3}{5}}.$$

Exercise 2.6(b)

Given further that the $|\phi_n\rangle$ are the eigenstates of an operator $\hat{B}$, with $\hat{B}|\phi_n\rangle = (3n^2 - 1)|\phi_n\rangle$, for $n \in \{1, 2, 3\}$.

By linearity, one can verify that

$$\begin{aligned}
 \hat{B}|\psi\rangle &= \hat{B}\left(\frac{1}{\sqrt{15}}|\phi_1\rangle + \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle\right) \\
 &= \frac{1}{\sqrt{15}}\hat{B}|\phi_1\rangle + \frac{1}{\sqrt{3}}\hat{B}|\phi_2\rangle + \frac{1}{\sqrt{5}}\hat{B}|\phi_3\rangle \\
 &= \frac{2}{\sqrt{15}}|\phi_1\rangle + \frac{11}{\sqrt{3}}|\phi_2\rangle + \frac{26}{\sqrt{5}}|\phi_3\rangle.
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \langle\psi|\hat{B}|\psi\rangle &= \left(\frac{1}{\sqrt{15}}\langle\phi_1| + \frac{1}{\sqrt{3}}\langle\phi_2| + \frac{1}{\sqrt{5}}\langle\phi_3|\right)\left(\frac{2}{\sqrt{15}}|\phi_1\rangle + \frac{11}{\sqrt{3}}|\phi_2\rangle + \frac{26}{\sqrt{5}}|\phi_3\rangle\right) \\
 &= \frac{2}{15} + \frac{11}{3} + \frac{26}{5} \\
 &= \frac{2 + 55 + 78}{15} \\
 &= \frac{135}{15} \\
 &= 9.
 \end{aligned}$$

The expectation value of $\hat{B}$ with respect to the state $|\psi\rangle$ is then

$$\begin{aligned}
 \langle\hat{B}\rangle &= \frac{\langle\psi|\hat{B}|\psi\rangle}{\langle\psi|\psi\rangle} \\
 &= \frac{9}{3/5} \\
 &= 15.
 \end{aligned}$$

where the second equality is a consequence of our work in part (a).

Exercise 2.6(c)

$$\begin{aligned}
 \hat{B}^2|\psi\rangle &= \hat{B}^2\left(\frac{1}{\sqrt{15}}|\phi_1\rangle + \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle\right) \\
 &= \frac{1}{\sqrt{15}}\hat{B}^2|\phi_1\rangle + \frac{1}{\sqrt{3}}\hat{B}^2|\phi_2\rangle + \frac{1}{\sqrt{5}}\hat{B}^2|\phi_3\rangle \\
 &= \frac{2^2}{\sqrt{15}}|\phi_1\rangle + \frac{11^2}{\sqrt{3}}|\phi_2\rangle + \frac{26^2}{\sqrt{5}}|\phi_3\rangle, \quad (\text{See Theorem A.1.}) \\
 &= \frac{4}{\sqrt{15}}|\phi_1\rangle + \frac{121}{\sqrt{3}}|\phi_2\rangle + \frac{676}{\sqrt{5}}|\phi_3\rangle.
 \end{aligned}$$

$$\begin{aligned}
\langle \psi | \hat{B}^2 | \psi \rangle &= \left(\frac{1}{\sqrt{15}} \langle \phi_1 | + \frac{1}{\sqrt{3}} \langle \phi_2 | + \frac{1}{\sqrt{5}} \langle \phi_3 | \right) \left(\frac{4}{\sqrt{15}} | \phi_1 \rangle + \frac{121}{\sqrt{3}} | \phi_2 \rangle + \frac{676}{\sqrt{5}} | \phi_3 \rangle \right) \\
&= \frac{4}{15} + \frac{121}{3} + \frac{676}{5} \\
&= \frac{4 + 605 + 2028}{15} \\
&= \frac{2637}{15} \\
&= \frac{879}{5}.
\end{aligned}$$

It follows that the expectation value of $\langle \hat{B}^2 \rangle$ is given by

$$\begin{aligned}
\langle \hat{B}^2 \rangle &= \frac{\langle \psi | \hat{B}^2 | \psi \rangle}{\langle \psi | \psi \rangle} \\
&= \frac{\left(\frac{879}{5} \right)}{3/5} \\
&= 293.
\end{aligned}$$

Exercises 2.7, 2.8, 2.9 and 2.10

All trivial.

For Exercise 2.7(b), you may wish to use Euler's identity, and for Exercise 2.8(e), the hyperbolic tangent identities may be useful. See, for instance, the formulae in [Tay05]

Exercise 2.11

Theorem. If $\hat{A}$ is projection operator, then $1 - \hat{A}$ is a projection operator.

Proof: Define the operator $\hat{B} = \hat{I} - \hat{A}$. As $\hat{A}$ is a projection operator by assumption, and by the properties of the adjoint, we have

$$\hat{B}^\dagger = (\hat{I} - \hat{A})^\dagger = \hat{I}^\dagger - \hat{A}^\dagger = \hat{I} - \hat{A} := \hat{B},$$

and

$$\hat{B}^2 = (\hat{I} - \hat{A})^2 = \hat{I} - 2\hat{A}\hat{I} + \hat{A}^2 = \hat{I} - 2\hat{A} + \hat{A} = \hat{I} - \hat{A} := \hat{B}$$

so that, by definition, $\hat{B}$ constitutes a projection operator, as required. ■

Exercise 2.1.2.

Define the operator

$$\hat{A} := \frac{|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle}.$$

We propose the following result:

Theorem. $\hat{A}$ is a projection operator.

Proof: Indeed, it is clear that $\hat{A}$ is self-adjoint. Now, observe

$$\begin{aligned}\hat{A}^2 &= \left(\frac{|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle} \right) \left(\frac{|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle} \right)^1 \\ &= \frac{|\psi\rangle\langle\psi|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle^2} \\ &= \frac{|\psi\rangle\langle\psi|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle^2} \\ &= \frac{|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle} \\ &= \hat{A},\end{aligned}$$

which verifies our claim². ■

Exercise 2.13(a)

For $\langle\phi|\hat{A}|\psi\rangle\langle\psi|$,

$$\begin{aligned}\left(\langle\phi|\hat{A}|\psi\rangle\langle\psi|\right)^\dagger &= \langle\psi|^\dagger \left(\langle\phi|\hat{A}|\psi\rangle\right)^\dagger \\ &= |\psi\rangle\langle\psi|\hat{A}^\dagger|\phi\rangle.\end{aligned}$$

Exercise 2.13(b)

Consider now $\hat{A}|\psi\rangle\langle\phi|$. It is easy to see that this acts as an operator, as for any state $|\chi\rangle$,

$$\left(\hat{A}|\psi\rangle\langle\phi|\right)|\chi\rangle = (\langle\phi|\chi\rangle)\hat{A}|\psi\rangle.$$

¹When evaluating T^n where T is some operator, *always* write, for example, $T^2 = TT$.

²Notice that we never assumed that the ket $|\psi\rangle$ was normalised.

$$\begin{aligned} (\hat{A}|\psi\rangle\langle\phi|)^\dagger &= \langle\phi|^\dagger|\psi\rangle^\dagger\hat{A}^\dagger \\ &= |\phi\rangle\langle\psi|\hat{A}^\dagger. \end{aligned}$$

Exercise 2.13(c)

$$\begin{aligned} (\langle\phi|\hat{A}|\psi\rangle|\psi\rangle\langle\phi|\hat{A})^\dagger &= \hat{A}^\dagger(|\psi\rangle\langle\phi|)^\dagger(\langle\phi|\hat{A}|\psi\rangle)^\dagger \\ &= (\hat{A}^\dagger|\phi\rangle\langle\psi|)(\langle\psi|\hat{A}^\dagger|\phi\rangle). \end{aligned}$$

Exercise 2.13(d)

$$\begin{aligned} (\langle\psi|\hat{A}|\phi\rangle|\phi\rangle + i\hat{A}|\psi\rangle)^\dagger &= (\langle\psi|\hat{A}|\phi\rangle|\phi\rangle)^\dagger + (i\hat{A}|\psi\rangle)^\dagger \\ &= \langle\phi|(\langle\psi|\hat{A}|\phi\rangle)^\dagger - i\langle\psi|\hat{A}^\dagger \\ &= \langle\phi|\langle\phi|\hat{A}^\dagger|\psi\rangle - i\langle\psi|\hat{A}^\dagger. \end{aligned}$$

Exercise 2.13(e)

$$\begin{aligned} (|\phi\rangle\langle\phi|\hat{A} - i\hat{A}|\psi\rangle\langle\psi|)^\dagger &= (|\phi\rangle\langle\phi|\hat{A})^\dagger + (-i\hat{A}|\psi\rangle\langle\psi|)^\dagger \\ &= \hat{A}^\dagger|\phi\rangle\langle\phi| + i|\psi\rangle\langle\psi|\hat{A}^\dagger. \end{aligned}$$

For proofs of the properties of adjoints, see 7.5 in [Axl24].

Exercise 2.14(a)

Since $|\phi_1\rangle$ and $|\phi_2\rangle$ are orthonormal, and is the same length as the dimension of the vector space (by assumption), we have that the list $|\phi_1\rangle, |\phi_2\rangle$ forms a basis by 6.28 in [Axl24].

Exercise 2.14(b)

Consider the operator $\hat{B} = |\phi_1\rangle\langle\phi_2|$. We have

$$\hat{B}^\dagger = |\phi_2\rangle\langle\phi_1| \neq \hat{B},$$

so $\hat{B}$ is not Hermitian.

Now, observe that

$$\begin{aligned}\hat{B}^2 &= \hat{B}\hat{B} = (|\phi_1\rangle\langle\phi_2|)(|\phi_1\rangle\langle\phi_2|) \\ &= |\phi_1\rangle\langle\phi_2||\phi_1\rangle\langle\phi_2| = 0,\end{aligned}$$

by orthonormality.

Exercise 2.14(c)

Theorem. The products $\hat{B}\hat{B}^\dagger$ and $\hat{B}^\dagger\hat{B}$ are projection operators.

Proof: By definition,

$$(\hat{B}\hat{B}^\dagger)^\dagger = (\hat{B}^\dagger)^\dagger\hat{B}^\dagger = \hat{B}\hat{B}^\dagger,$$

and

$$(\hat{B}^\dagger\hat{B})^\dagger = \hat{B}^\dagger(\hat{B}^\dagger)^\dagger = \hat{B}^\dagger\hat{B}.$$

Hence, $\hat{B}\hat{B}^\dagger$ and $\hat{B}^\dagger\hat{B}$ are both Hermitian.

It is easy to verify from the definition of $\hat{B}$ that $\hat{B}\hat{B}^\dagger = |\phi_1\rangle\langle\phi_1|$ and $\hat{B}^\dagger\hat{B} = |\phi_2\rangle\langle\phi_2|$ (do so now). It immediately follows that $(\hat{B}\hat{B}^\dagger)^2 = \hat{B}\hat{B}^\dagger$, and $(\hat{B}^\dagger\hat{B})^2 = \hat{B}^\dagger\hat{B}$ (why?) so that the two products constitute projection operators, as required. ■

Exercise 2.14(d)

Theorem. The operator $\hat{B}\hat{B}^\dagger - \hat{B}^\dagger\hat{B}$ is unitary.

Proof: Define $\hat{T} = \hat{B}\hat{B}^\dagger - \hat{B}^\dagger\hat{B}$. Our work above shows that

$$\hat{T} = |\phi_1\rangle\langle\phi_1| - |\phi_2\rangle\langle\phi_2|.$$

It follows that $\hat{T}|\phi_1\rangle = |\phi_1\rangle$ and $\hat{T}|\phi_2\rangle = -|\phi_2\rangle$. The matrix of $\hat{T}$ with respect to the orthonormal basis $|\phi_1\rangle, |\phi_2\rangle$ is, therefore

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

From 7.53 of [Axl24], we see that $\hat{T}$ is unitary, as required. ■

Exercise 2.14(e)

Trivial.

Exercise 2.16(a)

Consider the matrix

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

Its columns are given by $a_1 = (\cos \theta, \sin \theta)$ and $a_2 = (-\sin \theta, \cos \theta)$. With respect to the standard Euclidean norm, the Pythagorean identity shows that

$$\|a_1\| = \|a_2\| = 1,$$

and it is also easy to deduce that $\langle a_1, a_2 \rangle = 0$, i.e. the columns form an orthonormal list in $\mathbb{R}^2$, so A is a unitary matrix by 7.57(b) of [Ax124].

Exercise 2.16(b)

We treat the case $\theta \neq 0$, otherwise A is simply the identity.

To find the eigenvalues of A amounts to solving the equation $\det(A - \lambda I) = 0$ for λ . This is an easy computation, and we obtain

$$\lambda = e^{\pm i\theta},$$

to which we can find a pair of normalised eigenvectors

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \text{ and } v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix},$$

corresponding to the eigenvalues $\lambda_+ = e^{i\theta}$ and $\lambda_- = e^{-i\theta}$, respectively.

Remark: You might notice some similarity between A and the ordinary rotational matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The difference is that the matrix above represents a *counterclockwise* rotation about the origin, whereas A represents a *clockwise* rotation about the origin (see the figure below). That A has no real eigenvalues, except when $\theta = 0$, aligns with our intuition: Multiplication of a vector by a real scalar will not cause it to rotate!

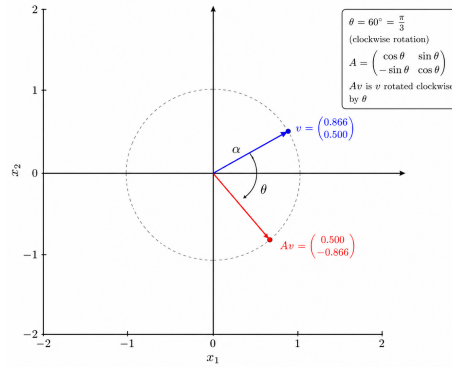


Figure 2.2: The action of A on a vector about the origin (image generated by ChatGPT)

Exercises 2.17, 2.18 and 2.19

All are computationally heavy, and you should be able to figure these out yourself by referring to the book when necessary.

Exercise 2.20

We claim to possess the following result:

Let $\hat{A}$, $\hat{B}$, and $\hat{C}$ be operators. Then we have

$$\text{Tr}(\hat{A}\hat{B}\hat{C}) = \text{Tr}(\hat{C}\hat{A}\hat{B}) = \text{Tr}(\hat{B}\hat{C}\hat{A}).$$

Proof: Note the identity $\text{Tr}(\hat{S}\hat{T}) = \text{Tr}(\hat{T}\hat{S})$. Observe that

$$\begin{aligned} \text{Tr}(\hat{A}\hat{B}\hat{C}) &= \text{Tr}((\hat{A}\hat{B})\hat{C}) = \text{Tr}(\hat{C}(\hat{A}\hat{B})) = \text{Tr}(\hat{C}\hat{A}\hat{B}) \\ &= \text{Tr}((\hat{C}\hat{A})\hat{B}) = \text{Tr}(\hat{B}(\hat{C}\hat{A})) = \text{Tr}(\hat{B}\hat{C}\hat{A}), \end{aligned}$$

which completes the proof. ■

Exercises 2.21, 2.22, 2.23, 2.24

All are computationally heavy, and you should be able to figure these out by referring to the book when necessary³.

³Unless you are a masochist, I would recommend using software to calculate the eigenvalues for you.

Exercise 2.25 Consider an operator $\hat{A}$ so that

$$[\hat{A}, \hat{A}^\dagger] = 1.$$

(a) Evaluate the commutators

$$[\hat{A}^\dagger \hat{A}, \hat{A}] \quad \text{and} \quad [\hat{A}^\dagger \hat{A}, \hat{A}^\dagger].$$

(b) If the actions of $\hat{A}$ and $\hat{A}^\dagger$ on the states $\{|a\rangle\}$ are given by

$$\hat{A}|a\rangle = \sqrt{a}|a-1\rangle$$

and

$$\hat{A}^\dagger|a\rangle = \sqrt{a+1}|a+1\rangle$$

and if $\langle a'|a\rangle = \delta_{a'a}$, calculate

$$\langle a|\hat{A}|a+1\rangle, \quad \langle a+1|\hat{A}^\dagger|a\rangle$$

and

$$\langle a|\hat{A}^\dagger \hat{A}|a\rangle, \quad \langle a|\hat{A} \hat{A}^\dagger|a\rangle.$$

(c) Calculate

$$\langle a|(\hat{A} + \hat{A}^\dagger)^2|a\rangle \quad \text{and} \quad \langle a|(\hat{A} - \hat{A}^\dagger)^2|a\rangle.$$

(a) Using the distributive properties of the commutator:

$$[\hat{A}^\dagger \hat{A}, \hat{A}] = \hat{A}^\dagger [\hat{A}, \hat{A}] + [\hat{A}^\dagger, \hat{A}] \hat{A} = -\hat{A}$$

where the last equality follows from anticommutativity.

For the second one:

$$[\hat{A}^\dagger \hat{A}, \hat{A}^\dagger] = \hat{A}^\dagger [\hat{A}, \hat{A}^\dagger] + [\hat{A}^\dagger, \hat{A}^\dagger] \hat{A} = \hat{A}^\dagger.$$

(b)

These are trivial calculations.

By definition,

$$\begin{aligned} \langle a|\hat{A}|a+1\rangle &= \langle a|\sqrt{a+1}|a\rangle \\ &= \sqrt{a+1}\langle a|a\rangle \\ &= \sqrt{a+1}. \end{aligned}$$

Similarly,

$$\begin{aligned}\langle a+1|\hat{A}^\dagger|a\rangle &= \langle a+1|\sqrt{a+1}|a+1\rangle \\ &= \sqrt{a+1}\langle a+1|a+1\rangle \\ &= \sqrt{a+1}.\end{aligned}$$

For the first product,

$$\begin{aligned}\langle a|\hat{A}^\dagger\hat{A}|a\rangle &= \langle a|\hat{A}^\dagger(\sqrt{a}|a-1\rangle) \\ &= \sqrt{a}\langle a|\hat{A}^\dagger|a-1\rangle \\ &= \sqrt{a}\langle a|\sqrt{a}|a\rangle \\ &= a\langle a|a\rangle \\ &= a.\end{aligned}$$

For the second product,

$$\begin{aligned}\langle a|\hat{A}\hat{A}^\dagger|a\rangle &= \langle a|\hat{A}(\sqrt{a+1}|a+1\rangle) \\ &= \sqrt{a+1}\langle a|\hat{A}|a+1\rangle \\ &= a+1.\end{aligned}$$

*Recall that the Kronecker Delta is defined as $\delta_{a,a'} = \begin{cases} 1, & a = a' \\ 0, & a \neq a'. \end{cases}$

(c)

Before directly attacking the problem, we may wish to evaluate

$$(\hat{A} + \hat{A}^\dagger)^2 = (\hat{A} + \hat{A}^\dagger)(\hat{A} + \hat{A}^\dagger) = \hat{A}^2 + \hat{A}\hat{A}^\dagger + \hat{A}^\dagger\hat{A} + (\hat{A}^\dagger)^2$$

4

$$(\hat{A} - \hat{A}^\dagger)^2 = (\hat{A} - \hat{A}^\dagger)(\hat{A} - \hat{A}^\dagger) = \hat{A}^2 - \hat{A}\hat{A}^\dagger - \hat{A}^\dagger\hat{A} + (\hat{A}^\dagger)^2$$

$$\begin{aligned}\langle a|(\hat{A} + \hat{A}^\dagger)^2|a\rangle &= \langle a|\left(\hat{A}^2 + \hat{A}\hat{A}^\dagger + \hat{A}^\dagger\hat{A} + (\hat{A}^\dagger)^2\right)|a\rangle \\ &= \langle a|\hat{A}^2|a\rangle + \langle a|\hat{A}\hat{A}^\dagger|a\rangle + \langle a|\hat{A}^\dagger\hat{A}|a\rangle + \langle a|(\hat{A}^\dagger)^2|a\rangle \\ &= 0 + (a+1) + a + 0 \\ &= 2a+1.\end{aligned}$$

⁴Why can't I combine $\hat{A}\hat{A}^\dagger + \hat{A}^\dagger\hat{A}$? Under what conditions can I do so?

I want you to think about why the first and last terms go to zero.

Hint: what are the properties of the Kronecker Delta?

Similarly,

$$\begin{aligned}
 \langle a | (\hat{A} - \hat{A}^\dagger)^2 | a \rangle &= \langle a | \left(\hat{A}^2 - \hat{A}\hat{A}^\dagger - \hat{A}^\dagger\hat{A} + (\hat{A}^\dagger)^2 \right) | a \rangle \\
 &= \langle a | \hat{A}^2 | a \rangle - \langle a | \hat{A}\hat{A}^\dagger | a \rangle - \langle a | \hat{A}^\dagger\hat{A} | a \rangle + \langle a | (\hat{A}^\dagger)^2 | a \rangle \\
 &= 0 - (a + 1) - a + 0 \\
 &= -2a - 1.
 \end{aligned}$$

Exercise 2.27 If $\hat{A}$ and $\hat{B}$ commute, and if $|\psi_1\rangle$ and $|\psi_2\rangle$ are two eigenvectors of $\hat{A}$ with different eigenvalues ($\hat{A}$ is Hermitian), show that

(a) $\langle \psi_1 | \hat{B} | \psi_2 \rangle$ is zero and

(b) $\hat{B}|\psi_1\rangle$ is also an eigenvector to $\hat{A}$ with the same eigenvalue as $|\psi_1\rangle$; i.e., if

$$\hat{A}|\psi_1\rangle = a_1|\psi_1\rangle,$$

show that

$$\hat{A}(\hat{B}|\psi_1\rangle) = a_1\hat{B}|\psi_1\rangle.$$

Solution

Exercise 2.28 Let A and B be two $n \times n$ matrices. Assuming that B^{-1} exists, show that

$$[A, B^{-1}] = -B^{-1}[A, B]B^{-1}.$$

Proof:

$$\begin{aligned}
 -B^{-1}[A, B]B^{-1} &= -B^{-1}(AB - BA)B^{-1} = -B^{-1}(A - BAB^{-1}) \\
 &= AB^{-1} - B^{-1}A := [A, B^{-1}].
 \end{aligned}$$

Exercise 2.33 Show that

$$[\hat{X}, \hat{P}^n] = i\hbar n\hat{P}^{n-1}.$$

Proof:

We have the relation $[\hat{X}, \hat{P}] = i\hbar$. Therefore,

$$\begin{aligned}
 [\hat{X}, \hat{P}^n] &= \sum_{j=0}^{n-1} \hat{P}^j [\hat{X}, \hat{P}] \hat{P}^{n-j-1} = i\hbar \left(\sum_{j=0}^{n-1} \hat{P}^{n-1} \right) \\
 &= i\hbar \left(\sum_{j=0}^{n-1} 1 \right) \hat{P}^{n-1} \\
 &= i\hbar n \hat{P}^{n-1}
 \end{aligned}$$

which completes the proof. ■

Chapter 3

Postulates of Quantum Mechanics

Brief note

Before you read my solutions, note that the functions

$$\phi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi n x}{a}\right) \quad (3.1)$$

are orthonormal.^a We also have the identity

$$\frac{d^2}{dx^2} \phi_n(x) = -\frac{n^2 \pi^2}{a^2} \phi_n(x). \quad (3.2)$$

^aThis is easy to verify; see Exercise 5 of Chapter 1 in [SS03]

Exercise 3.1

A particle in an infinite potential box with walls at $x = 0$ and $x = a$ (i.e., the potential is infinite for $x < 0$ and $x > a$ and zero in between) has the following wave function at some initial time:

$$\psi(x) = \frac{1}{\sqrt{5a}} \sin\left(\frac{\pi x}{a}\right) + \frac{2}{\sqrt{5a}} \sin\left(\frac{3\pi x}{a}\right).$$

- (a) Find the possible results of the measurement of the system's energy and the corresponding probabilities.
- (b) Find the form of the wave function after such a measurement.
- (c) If the energy is measured again immediately afterwards, what are the relative probabilities of the possible outcomes?

Solution to (a)

It is convenient to write our wave function ψ in terms of the ϕ_n ; all that is needed is to multiply each ϕ_n by a factor of $\sqrt{\frac{a}{2}}$:

$$\begin{aligned}\psi(x) &= \frac{1}{\sqrt{5a}} \sin\left(\frac{\pi x}{a}\right) + \frac{2}{\sqrt{5a}} \sin\left(\frac{3\pi x}{a}\right) \\ &= \frac{1}{\sqrt{10}} \phi_1(x) + \sqrt{\frac{2}{5}} \phi_3(x).\end{aligned}$$

Note that ψ is not normalised; by Parseval's Identity we have

$$\langle \psi | \psi \rangle = \frac{1}{10} + \frac{2}{5} = \frac{1}{2}.$$

In general, a system's Hamiltonian is given by

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V(\hat{R}, t),$$

where ∇^2 denotes the Laplacian. In the one-dimensional case, and more specifically this exercise, it simplifies to¹

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}.$$

The expectation value of the Hamiltonian².

$$\begin{aligned}\langle \hat{H} \rangle &= \langle \phi_n | \hat{H} | \phi_n \rangle \\ &= -\frac{\hbar^2}{2m} \int_0^a \phi_n^*(x) \frac{d^2}{dx^2} \phi_n(x) dx = \frac{n^2 \hbar^2 \pi^2}{2ma^2},\end{aligned}$$

so we obtain the energies

$$E_n = \frac{\langle \phi_n | \hat{H} | \phi_n \rangle}{\langle \psi | \psi \rangle} = 2 \langle \phi_n | \hat{H} | \phi_n \rangle = \frac{n^2 \hbar^2 \pi^2}{ma^2}.$$

¹Why does the potential vanish?

²For an alternative derivation of the corresponding energies of ϕ_n , you can read my article [Eis26]

The corresponding probabilities are

$$P_n(E_n) = \frac{|\langle \phi_n | \psi \rangle|^2}{\langle \psi | \psi \rangle} = 2|\langle \phi_n | \psi \rangle|^2.$$

That $P_1(E_1) = \frac{1}{5}$ and $P_3(E_3) = \frac{4}{5}$ is a verification left to the reader.

Hence, the possible outcomes of a measurement of the system's energy along with the corresponding probabilities are

$$\begin{aligned} E_1 &= \frac{\pi^2 \hbar^2}{ma^2}, & P_1(E_1) &= \frac{1}{5}, \\ E_3 &= \frac{9\pi^2 \hbar^2}{ma^2}, & P_3(E_3) &= \frac{4}{5}. \end{aligned}$$

Solution to (b)

The wave function $\psi(x, t)$ at any later time t is given by

$$\psi(x, t) = \frac{1}{\sqrt{10}}\phi_1(x)e^{-iE_1t/\hbar} + \sqrt{\frac{2}{5}}\phi_3(x)e^{-iE_3t/\hbar},$$

where the energies E_1 and E_3 are listed above.

Solution to (c)

The relative probabilities of the possible measurement outcomes of the system's energy remain constant; see the discussion of measurement on page 195 of the text.

Exercise 3.3

Consider a neutron which is confined to an infinite potential well of width $a = 8 \text{ fm}$. At time $t = 0$ the neutron is assumed to be in the state

$$\psi(x, 0) = \sqrt{\frac{4}{7a}} \sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{2}{7a}} \sin\left(\frac{2\pi x}{a}\right) + \sqrt{\frac{8}{7a}} \sin\left(\frac{3\pi x}{a}\right).$$

- If an energy measurement is carried out on the system, what are the values that will be found for the energy and with what probabilities? Express your answer in MeV (the mass of the neutron is $mc^2 = 939 \text{ MeV}$, $\hbar c = 197 \text{ MeV fm}$).
- If this measurement is repeated on many identical systems, what is the average value of the energy that will be found? Again, express your answer in MeV.
- Using the uncertainty principle, estimate the order of magnitude of the neutron's speed in this well as a function of the speed of light c .

Solution to (a)

Observe that

$$\begin{aligned} \psi(x, 0) &= \sqrt{\frac{4}{7a}} \sin\left(\frac{\pi x}{a}\right) + \sqrt{\frac{2}{7a}} \sin\left(\frac{2\pi x}{a}\right) + \sqrt{\frac{8}{7a}} \sin\left(\frac{3\pi x}{a}\right) \\ &= \sqrt{\frac{a}{2}} \left(\sqrt{\frac{4}{7a}} \phi_1(x) + \sqrt{\frac{2}{7a}} \phi_2(x) + \sqrt{\frac{8}{7a}} \phi_3(x) \right) \\ &= \sqrt{\frac{2}{7}} \phi_1(x) + \sqrt{\frac{1}{7}} \phi_2(x) + \sqrt{\frac{4}{7}} \phi_3(x). \end{aligned}$$

The expectation value of the Hamiltonian gives the energies³

$$\begin{aligned} E_n &= \langle \phi_n | \hat{H} | \phi_n \rangle \\ &= \frac{n^2 \hbar^2 \pi^2}{2ma^2}. \end{aligned}$$

The corresponding probability of E_n being measured is $P_n(E_n) = |\langle \phi_n | \psi \rangle|^2$. It is easy to verify that

³It is in your interest to verify that ψ is normalised; compare the formula for E_n above with the one I employ in Exercise 3.1.

$$E_1 = \langle \phi_1 | \hat{H} | \phi_1 \rangle = \frac{\pi^2 \hbar^2}{2ma^2}, \quad P_1(E_1) = |\langle \phi_1 | \psi \rangle|^2 = \frac{2}{7},$$

$$E_2 = \langle \phi_2 | \hat{H} | \phi_2 \rangle = \frac{4\pi^2 \hbar^2}{2ma^2}, \quad P_2(E_2) = |\langle \phi_2 | \psi \rangle|^2 = \frac{1}{7},$$

$$E_3 = \langle \phi_3 | \hat{H} | \phi_3 \rangle = \frac{9\pi^2 \hbar^2}{2ma^2}, \quad P_3(E_3) = |\langle \phi_3 | \psi \rangle|^2 = \frac{4}{7}.$$

Solution to (b)

The average energy is given by

$$\begin{aligned} E_{\text{avg}} &= \sum_n P_n E_n = \frac{2}{7} E_1 + \frac{1}{7} E_2 + \frac{4}{7} E_3 \\ &= \frac{3\hbar^2 \pi^2}{ma^2}. \end{aligned}$$

We may rewrite this as

$$\begin{aligned} E_{\text{avg}} &= \frac{3\hbar^2 c^2 \pi^2}{mc^2 a^2} = \frac{3\pi^2 (197 \text{ MeV fm})^2}{(939 \text{ MeV})(8 \text{ fm})^2} \\ &= 19.1 \text{ MeV (3s.f.)}. \end{aligned}$$

Solution to (c)

Heisenberg's Uncertainty Principle implies

$$\Delta p \Delta x \geq \frac{\hbar}{2}$$

Rearranging, it follows that

$$\Delta v \geq \frac{\hbar}{2ma} = 3.93 \times 10^6 \text{ m s}^{-1} \text{ (3s.f.)}.$$

Exercise 3.4

Consider the dimensionless harmonic oscillator Hamiltonian

$$\hat{H} = \frac{1}{2}\hat{P}^2 + \frac{1}{2}\hat{X}^2 \quad \text{with} \quad \hat{P} = -i\frac{d}{dx}.$$

- (a) Show that the two wave functions $\psi_0(x) = e^{-x^2/2}$ and $\psi_1(x) = xe^{-x^2/2}$ are eigenfunctions of H with eigenvalues $1/2$ and $3/2$, respectively.
- (b) Find the value of the coefficient α such that

$$\psi_2(x) = (1 + \alpha x^2)e^{-x^2/2}$$

is orthogonal to $\psi_0(x)$. Then show that $\psi_2(x)$ is an eigenfunction of H with eigenvalue $5/2$.

Solution to (a)

To make our lives easier, let's first compute $\hat{P}^2$:

$$\begin{aligned} \hat{P}^2 &= \left(-i\frac{d}{dx}\right)\left(-i\frac{d}{dx}\right) \\ &= i^2 \frac{d^2}{dx^2} \\ &= -\frac{d^2}{dx^2}. \end{aligned}$$

Our Hamiltonian is therefore given by

$$\hat{H} = -\frac{1}{2}\frac{d^2}{dx^2} + \frac{1}{2}\hat{X}^2.$$

We are now in a good position to attack the problem succinctly.

Proof that ψ_0 is an eigenfunction:

$$\begin{aligned}
 \hat{H}\psi_0(x) &= \left[-\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2} x^2 \right] \psi_0(x) \\
 &= -\frac{1}{2} \frac{d^2}{dx^2} \psi_0(x) + \frac{1}{2} x^2 \psi_0(x) \\
 &= -\frac{1}{2} \frac{d}{dx} \left[-x e^{-x^2/2} \right] + \frac{1}{2} x^2 \psi_0(x) \\
 &= -\frac{1}{2} \left(-\psi_0(x) + x^2 \psi_0(x) \right) + \frac{1}{2} x^2 \psi_0(x) \\
 &= \frac{1}{2} e^{-x^2/2} \\
 &= \frac{1}{2} \psi_0(x).
 \end{aligned}$$

In particular, we have $\hat{H}\psi_0 = \frac{1}{2}\psi_0$, so ψ_0 is an eigenfunction of the Hamiltonian with corresponding eigenvalue $\frac{1}{2}$ as claimed. ■

Proof that ψ_1 is an eigenfunction:

$$\begin{aligned}
 \hat{H}\psi_1(x) &= \left[-\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2} x^2 \right] \psi_1(x) \\
 &= -\frac{1}{2} \frac{d}{dx} \left[e^{-x^2/2} - x^2 e^{-x^2/2} \right] + \frac{1}{2} x^2 \psi_1(x) \\
 &= -\frac{1}{2} \left(-x e^{-x^2/2} - 2x e^{-x^2/2} + x^3 e^{-x^2/2} \right) + \frac{1}{2} x^2 \psi_1(x) \\
 &= \frac{1}{2} (3\psi_1(x) - x^2 \psi_1(x)) + \frac{1}{2} x^2 \psi_1(x) \\
 &= \frac{3}{2} \psi_1(x).
 \end{aligned}$$

In particular, we have $\hat{H}\psi_1 = \frac{3}{2}\psi_1$, so ψ_1 is an eigenfunction of the Hamiltonian with corresponding eigenvalue $\frac{3}{2}$. ■

Solution to (b)

In order for ψ_2 to be orthogonal to ψ_0 , we must require that $\langle \psi_0 | \psi_2 \rangle = 0$. Now, note

$$\begin{aligned}
 \langle \psi_0 | \psi_2 \rangle &= \int_{\mathbb{R}} \psi_0^*(x) \psi_2(x) dx \\
 &= \int_{\mathbb{R}} e^{-x^2/2} (1 + \alpha x^2) e^{-x^2/2} dx = \int_{\mathbb{R}} (1 + \alpha x^2) e^{-x^2} dx \\
 &= \int_{\mathbb{R}} e^{-x^2} dx + \alpha \int_{\mathbb{R}} x^2 e^{-x^2} dx \\
 &= \sqrt{\pi} + 2\alpha \int_0^{\infty} x^2 e^{-x^2} dx.
 \end{aligned}$$

Consider the substitution $u = x^2$ in the second integral. Clearly the limits of the integral (at 0 and ∞) remain the same, and $du = 2x dx = 2\sqrt{u} dx$ by definition. We therefore have⁴

$$\begin{aligned}
 \langle \psi_0 | \psi_2 \rangle &= \sqrt{\pi} + 2\alpha \int_0^{\infty} u e^{-u} \frac{du}{2\sqrt{u}} \\
 &= \sqrt{\pi} + \alpha \int_0^{\infty} u^{1/2} e^{-u} du \\
 &= \sqrt{\pi} + \alpha \int_0^{\infty} u^{\frac{3}{2}-1} e^{-u} du \\
 &= \sqrt{\pi} + \alpha \Gamma\left(\frac{3}{2}\right) \\
 &= \left(1 + \frac{\alpha}{2}\right) \sqrt{\pi}.
 \end{aligned}$$

Invoking our condition $\langle \psi_0 | \psi_2 \rangle = 0$ forces $\alpha = -2$ from the expression above.

It follows that ψ_2 is given by the equation

$$\psi_2(x) = (1 - 2x^2)e^{-x^2/2},$$

so that

⁴If you do not know that the symbol Γ is in the equation below, see my remark at the end of this solution.

$$\begin{aligned}
\hat{H}\psi_2(x) &= \left[-\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2} x^2 \right] \psi_2(x) \\
&= -\frac{1}{2} \frac{d}{dx} \left[(-4x - x + 2x^3) e^{-x^2/2} \right] + \frac{1}{2} x^2 \psi_2(x) \\
&= -\frac{1}{2} \frac{d}{dx} \left[(2x^3 - 5x) e^{-x^2/2} \right] + \frac{1}{2} x^2 \psi_2(x) \\
&= -\frac{1}{2} \left[(6x^2 - 5) e^{-x^2/2} - x(2x^3 - 5x) e^{-x^2/2} \right] + \frac{1}{2} x^2 \psi_2(x) \\
&= -\frac{1}{2} (6x^2 - 5 - 2x^4 + 5x^2) e^{-x^2/2} + \frac{1}{2} x^2 \psi_2(x) \\
&= \frac{1}{2} \left(5(2 - 2x^2) e^{-x^2/2} - x^2(1 - 2x^2) e^{-x^2/2} \right) + \frac{1}{2} x^2 \psi_2(x) \\
&= \frac{1}{2} (5\psi_2(x) - x^2\psi_2(x)) - \frac{1}{2} x^2 \psi_2(x) \\
&= \frac{5}{2} \psi_2(x).
\end{aligned}$$

Therefore, ψ_2 is an eigenfunction of the Hamiltonian with corresponding eigenvalue $\frac{5}{2}$, as was to be shown.

Remark

In case you were not aware, Γ denotes the Gamma function, which is defined by the integral

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

This function is a generalisation of the factorial function to the complex numbers, and has a few interesting properties. One of them is

$$\Gamma(z+1) = z\Gamma(z).$$

More importantly, one can show^a

$$\Gamma\left(\frac{3}{2}\right) = \sqrt{\frac{\pi}{2}},$$

which is the result used in the solution.

I have included a diagram of the Gamma function below, for the sole reason that it is nice to look at.

^aFor more intricate discussions about the Gamma function, such as the convergence of the integral, along with more identities, see [SS03]

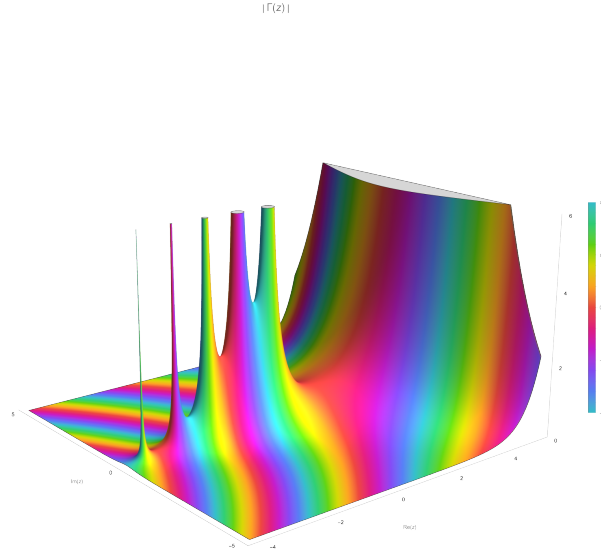


Figure 3.1: Plot of the Gamma function over the complex plane. Image from [Wikipedia](#).

Exercise 3.5

Consider that the wave function of a dimensionless harmonic oscillator, whose Hamiltonian is

$$\hat{H} = \frac{1}{2}\hat{P}^2 + \frac{1}{2}\hat{X}^2,$$

is given at time $t = 0$ by

$$\psi(x, 0) = \frac{1}{\sqrt{8\pi}}\phi_0(x) + \frac{1}{\sqrt{18\pi}}\phi_2(x) = \frac{1}{\sqrt{8\pi}}e^{-x^2/2} + \frac{1}{\sqrt{18\pi}}(1-2x^2)e^{-x^2/2}.$$

- Find the expression of the oscillator's wave function at any later time t .
- Calculate the probability P_0 to find the system in an eigenstate of energy $1/2$ and the probability P_2 of finding the system in an eigenstate of energy $5/2$.
- Calculate the probability density, $\rho(x, t)$, and the current density, $\mathbf{J}(x, t)$.
- Verify that the probability is conserved; that is, show that

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J}(x, t) = 0.$$

Exercise 3.6 A particle of mass m , in an infinite potential well of length a , has the following initial wave function at $t = 0$:

$$\psi(x, 0) = \sqrt{\frac{3}{5a}} \sin\left(\frac{3\pi x}{a}\right) + \frac{1}{\sqrt{5a}} \sin\left(\frac{5\pi x}{a}\right),$$

and an energy spectrum

$$E_n = \frac{\hbar^2 \pi^2 n^2}{2ma^2}.$$

Find $\psi(x, t)$ at any later time t , then calculate $\partial \rho / \partial t$ and the probability current density vector $\mathbf{J}(x, t)$ and verify that

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J}(x, t) = 0.$$

Recall that

$$\rho = \psi^*(x, t)\psi(x, t)$$

and

$$\mathbf{J}(x, t) = \frac{i\hbar}{2m} [\psi(x, t)\nabla\psi^*(x, t) - \psi^*(x, t)\nabla\psi(x, t)].$$

Solution

We may write the initial wave function as

$$\begin{aligned}\psi(x, 0) &= \sqrt{\frac{3}{5a}} \sqrt{\frac{a}{2}} \phi_3(x) + \frac{1}{\sqrt{5a}} \sqrt{\frac{a}{2}} \phi_5(x) \\ &= \sqrt{\frac{3}{10}} \phi_3(x) + \sqrt{\frac{1}{10}} \phi_5(x).\end{aligned}$$

The corresponding energy levels to ϕ_3 and ϕ_5 are given by substituting $n = 3$ and $n = 5$ into the energy spectrum:

$$E_3 = \frac{9\pi^2\hbar^2}{2ma^2} \qquad E_5 = \frac{25\pi^2\hbar^2}{2ma^2}.$$

Therefore, the wave function at any later time t $\psi(x, t)$ is given by

$$\psi(x, t) = \sqrt{\frac{3}{10}} \phi_3(x) e^{-iE_3 t/\hbar} + \sqrt{\frac{1}{10}} \phi_5(x) e^{-iE_5 t/\hbar}.$$

Since we are working in $\mathbb{R}^1$, the probability density and current density vector are given by the equations

$$\rho(x, t) = \psi^*(x, t) \psi(x, t) \qquad \vec{J}(x, t) = \frac{i\hbar}{2m} \left(\psi \frac{\partial \psi^*}{\partial x} - \psi^* \frac{\partial \psi}{\partial x} \right) \hat{x}$$

For the probability density, we take the conjugate:

$$\begin{aligned}\rho(x, t) &= \psi^*(x, t) \psi(x, t) \\ &= \left(\sqrt{\frac{3}{10}} \phi_3 e^{iE_3 t/\hbar} + \sqrt{\frac{1}{10}} \phi_5 e^{iE_5 t/\hbar} \right) \left(\sqrt{\frac{3}{10}} \phi_3 e^{-iE_3 t/\hbar} + \sqrt{\frac{1}{10}} \phi_5 e^{-iE_5 t/\hbar} \right) \\ &= \frac{3}{10} \phi_3^2 + \frac{1}{10} \phi_5^2 + \frac{\sqrt{3}}{10} \phi_3 \phi_5 \left[e^{i(E_3 - E_5)t/\hbar} + e^{i(E_5 - E_3)t/\hbar} \right] \\ &= \frac{3}{10} \phi_3^2(x) + \frac{1}{10} \phi_5^2(x) + \frac{\sqrt{3}}{5} \phi_3(x) \phi_5(x) \cos \left(\frac{E_5 - E_3}{\hbar} t \right).\end{aligned}$$

$$\begin{aligned}\frac{\partial \psi}{\partial x} &= \sqrt{\frac{3}{10}} \phi_3'(x) e^{-iE_3 t/\hbar} + \sqrt{\frac{1}{10}} \phi_5'(x) e^{-iE_5 t/\hbar}, \\ \frac{\partial \psi^*}{\partial x} &= \sqrt{\frac{3}{10}} \phi_3'(x) e^{iE_3 t/\hbar} + \sqrt{\frac{1}{10}} \phi_5'(x) e^{iE_5 t/\hbar}.\end{aligned}$$

For the current density vector, we first compute the partial derivatives:

$$\begin{aligned}\frac{\partial\psi}{\partial x} &= \sqrt{\frac{3}{10}}\phi'_3(x)e^{-iE_3t/\hbar} + \sqrt{\frac{1}{10}}\phi'_5(x)e^{-iE_5t/\hbar}, \\ \frac{\partial\psi^*}{\partial x} &= \sqrt{\frac{3}{10}}\phi'_3(x)e^{iE_3t/\hbar} + \sqrt{\frac{1}{10}}\phi'_5(x)e^{iE_5t/\hbar}.\end{aligned}$$

$$\begin{aligned}\vec{J}(x, t) &= \frac{i\hbar}{2m} \left(\psi \frac{\partial\psi^*}{\partial x} - \psi^* \frac{\partial\psi}{\partial x} \right) \hat{\mathbf{x}} \\ &= -\frac{\sqrt{3}\hbar}{10m} [\phi_3(x)\phi'_5(x) - \phi_5(x)\phi'_3(x)] \sin\left(\frac{E_5 - E_3}{\hbar}t\right) \hat{\mathbf{x}}.\end{aligned}$$

Finally, we verify the Continuity Equation:

$$\frac{\partial\rho}{\partial t} + \nabla \cdot \vec{J} = 0$$

Since $\vec{J} = J_x \hat{\mathbf{x}}$,

$$\nabla \cdot \vec{J} = \frac{\partial J_x}{\partial x}.$$

From the expression for $\rho(x, t)$ determined above,

$$\frac{\partial\rho}{\partial t} = -\frac{\sqrt{3}}{5} \frac{E_5 - E_3}{\hbar} \phi_3\phi_5 \sin\left(\frac{E_5 - E_3}{\hbar}t\right).$$

For the current,

$$\begin{aligned}\frac{\partial J_x}{\partial x} &= -\frac{\sqrt{3}\hbar}{10m} \frac{\partial}{\partial x} (\phi_3\phi'_5 - \phi_5\phi'_3) \sin\left(\frac{E_5 - E_3}{\hbar}t\right) \\ &= -\frac{\sqrt{3}\hbar}{10m} (\phi_3\phi''_5 - \phi_5\phi''_3) \sin\left(\frac{E_5 - E_3}{\hbar}t\right).\end{aligned}$$

From the Remark at the beginning of this Chapter⁵, we have⁶

$$\phi''_n(x) = -\frac{2mE_n}{\hbar^2} \phi_n(x).$$

⁵Rearrange terms in the energy spectrum to make $\pi^2 n^2$ the subject.

⁶One can also use the time-independent Schrödinger Equation to obtain the result.

Therefore,

$$\begin{aligned}\phi_3\phi_5'' - \phi_5\phi_3'' &= -\frac{2mE_5}{\hbar^2}\phi_3\phi_5 + \frac{2mE_3}{\hbar^2}\phi_5\phi_3 \\ &= -\frac{2m(E_5 - E_3)}{\hbar^2}\phi_3\phi_5.\end{aligned}$$

Hence,

$$\frac{\partial J_x}{\partial x} = \frac{\sqrt{3}}{5} \frac{E_5 - E_3}{\hbar} \phi_3\phi_5 \sin\left(\frac{E_5 - E_3}{\hbar}t\right).$$

Finally, since the divergence in $\mathbb{R}^1$ is simply equal to $\partial/\partial x$, the Continuity Equation is satisfied:

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} &= -\frac{\sqrt{3}}{5} \frac{E_5 - E_3}{\hbar} \phi_3\phi_5 \sin\left(\frac{E_5 - E_3}{\hbar}t\right) \\ &\quad + \frac{\sqrt{3}}{5} \frac{E_5 - E_3}{\hbar} \phi_3\phi_5 \sin\left(\frac{E_5 - E_3}{\hbar}t\right) \\ &= 0.\end{aligned}$$

Exercise 3.7 Consider a system whose initial state at $t = 0$ is given in terms of a complete and orthonormal set of three vectors $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ as follows:

$$|\psi(0)\rangle = \frac{1}{\sqrt{3}}|\phi_1\rangle + A|\phi_2\rangle + \frac{1}{\sqrt{6}}|\phi_3\rangle,$$

where A is a real constant.

- Find A so that $|\psi(0)\rangle$ is normalized.
- If the energies corresponding to $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ are given by E_1 , E_2 , and E_3 , respectively, write down the state of the system $|\psi(t)\rangle$ at any later time t .
- Determine the probability of finding the system at a time t in the state $|\phi_3\rangle$.

Solution to (a)

By Parseval's Identity⁷

$$\langle\psi(0)|\psi(0)\rangle = \frac{1}{3} + A^2 + \frac{1}{6};$$

⁷See 6.30(b) in [Axl24].

for normalisation, we require that the equation above equals 1, so that

$$A = \frac{1}{\sqrt{2}}.$$

Solution to (b)

The state of the system at any later time t will be given by

$$|\psi(0)\rangle = \frac{1}{\sqrt{3}}e^{-iE_1t/\hbar}|\phi_1\rangle + \frac{1}{\sqrt{2}}e^{-iE_2t/\hbar}|\phi_2\rangle + \frac{1}{\sqrt{6}}e^{-iE_3t/\hbar}|\phi_3\rangle$$

Solution to (c)

The probability P of finding the system in this state is $P = |\langle\phi_3|\psi\rangle|^2$, so by orthonormality,

$$\begin{aligned} P &= \frac{1}{6} \\ &= 0.167 \text{ (3s.f.)}. \end{aligned}$$

Exercise 3.10

A particle bound in a one-dimensional potential has a wave function

$$\psi(x) = \begin{cases} Ae^{5ikx} \cos\left(\frac{3\pi x}{a}\right), & -\frac{a}{2} \leq x \leq \frac{a}{2}, \\ 0, & x < -\frac{a}{2} \text{ or } x > \frac{a}{2}. \end{cases}$$

- (a) Calculate the constant A so that $\psi(x)$ is normalized.
- (b) Calculate the probability of finding the particle between $x = 0$ and $x = a/4$.

Solution to (a)

For ψ to be normalised, require $\int_{\mathbb{R}} |\psi(x)|^2 dx = 1$. That is, set

$$\begin{aligned}
1 &= \int_{-\infty}^{\infty} |\psi(x)|^2 dx = \int_{-a/2}^{a/2} A^2 \cos^2 \left(\frac{3\pi x}{a} \right) dx \\
&= A^2 \int_{-a/2}^{a/2} \frac{1}{2} + \frac{1}{2} \cos \left(\frac{6\pi x}{a} \right) dx \\
&= A^2 \left[\frac{x}{2} + \frac{a}{12\pi} \sin \left(\frac{6\pi x}{a} \right) \right]_{-a/2}^{a/2} = \frac{A^2 a}{2},
\end{aligned}$$

which is to say

$$A = \sqrt{\frac{2}{a}}.$$

Solution to (b)

From part (a), our normalised wave function is given by

$$\psi(x) = \sqrt{\frac{2}{a}} e^{5ikx} \cos \left(\frac{3\pi x}{a} \right),$$

in the interval $\left[-\frac{a}{2}, \frac{a}{2}\right]$.

The Born Rule implies that the probability of finding the particle between $x = 0$ and $x = \frac{a}{4}$ is

$$\begin{aligned}
P &= \int_0^{a/4} |\psi(x)|^2 dx = \frac{2}{a} \int_0^{a/4} \cos^2 \left(\frac{3\pi x}{a} \right) dx \\
&= \frac{2}{a} \left[\frac{x}{2} + \frac{a}{12\pi} \sin \left(\frac{6\pi x}{a} \right) \right]_0^{a/4} = \frac{1}{4} - \frac{1}{6\pi} \\
&= 0.197 \text{ (3s.f.)}.
\end{aligned}$$

Exercise 3.13

Show that the momentum and the total energy can be measured simultaneously only when the potential is constant everywhere.

Solution

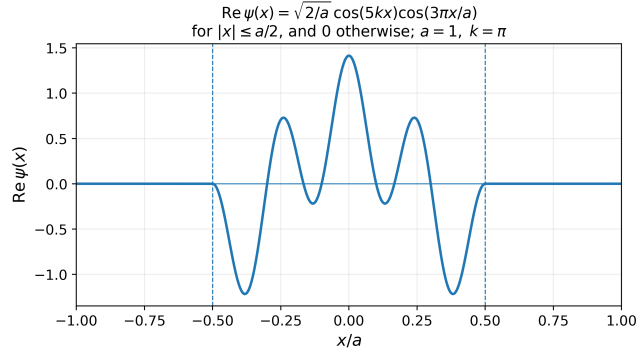


Figure 3.2: Plot of the real part of $\psi(x)$, for parameters $a = 1$ and $k = \pi$. Image generated by ChatGPT 5.6 Luna.

This follows immediately from Exercise 3.11(b); the momentum operator is compatible with the Hamiltonian operator iff the potential operator is constant in spatial coordinates.

Exercise 3.15 The complete set expansion of an initial wave function $\psi(x, 0)$ of a system in terms of orthonormal energy eigenfunctions $\phi_n(x)$ of the system has three terms, $n = 1, 2, 3$. The measurement of energy on the system represented by $\psi(x, 0)$ gives three values, E_1 and E_2 with probability $1/4$ and E_3 with probability $1/2$.

- Write down $\psi(x, 0)$ in terms of $\phi_1(x)$, $\phi_2(x)$, and $\phi_3(x)$.
- Find $\psi(x, 0)$ at any later time t , i.e., find $\psi(x, t)$.

Solution to (a)

$\psi(x, 0)$ is simply a linear combination of the ϕ_k 's:

$$\psi(x, 0) = a_1\phi_1(x) + a_2\phi_2(x) + a_3\phi_3(x).$$

Assuming that $\psi(x, 0)$ is normalised, the probability the energy is measured as E_k is $|\langle\phi_k|\psi\rangle|^2$. It follows that⁸

$$a_1^2 = \frac{1}{4}, a_2^2 = \frac{1}{4}, \text{ and } a_3^2 = \frac{1}{2}.$$

We may therefore write the wave function as

$$\psi(x, 0) = \frac{1}{2}\phi_1(x) + \frac{1}{2}\phi_2(x) + \frac{1}{\sqrt{2}}\phi_3(x).$$

⁸We assume that the a_k are purely real for simplicity.

Solution to (b)

It immediately follows from part (a) that the wave function at any later time t , $\psi(x, t)$, is given by

$$\psi(x, t) = \frac{1}{2}e^{-iE_1/\hbar}\phi_1(x) + \frac{1}{2}e^{-iE_2/\hbar}\phi_2(x) + \frac{1}{\sqrt{2}}e^{-iE_3/\hbar}\phi_3(x).$$

Appendix A

For the following result, I use the same notation as [Axl24].

Theorem A.1. Suppose $T \in \mathcal{L}(V)$, and v is an eigenvector of T corresponding to the eigenvalue λ , that is, $v \in V$ is such that $Tv = \lambda v$. Then for $n \in \mathbb{Z}^+$, we have $T^n v = \lambda^n v$.

Proof: The proof is by induction. Clearly the result is true for $n = 1$, so for our inductive step, suppose that this holds for some $n = k \geq 1$. Then, for $n = k + 1$, we have, by linearity,

$$T^n v = T(T^k v) = T(\lambda^k v) = \lambda^k T v = \lambda^k (\lambda v) = \lambda^{k+1} v = \lambda^n v,$$

which completes the proof. ■

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